

SUPPLEMENT TO “IDENTIFYING RATIONAL TYPES IN UNKNOWN
ENVIRONMENTS”
APPENDIX 4: STREAMLIT CALIBRATION MODEL

HUMBERTO BERNAL

Economics Program, Universidad Colegio Mayor de Cundinamarca and Ph.D. in Economics, Universidad de los Andes

S1. PURPOSE OF THE APPENDIX

This appendix documents the calibrated Streamlit model distributed in `../Streamlit/app.py`. The purpose is not to describe the graphical interface, but to state the economic objects computed by the application, the inputs required to run them, and the outputs generated for the empirical calibration, Table 5.2, and the dynamic diagnostics in Tab 6.

The application implements a dynamic kidnapping environment in which a criminal organization, the family, and the state choose intended actions. These intended actions are transformed into executed actions by an implementation kernel. The executed profile determines competing risks of termination, public signals, and Bayesian updating over the latent criminal type.

S2. REPLICATION PACKAGE

The folder `../Streamlit` contains the full replication package for the calibration app. The main entry point is `../Streamlit/app.py`. The main supporting files are `../Streamlit/model_logic.py`, which contains the structural hazard and payoff calculations, and `../Streamlit/rational_behavior.py`, which contains the behavioral, Bayesian, and mechanism calculations used by the app.

The package is run locally with

```
python3 -m streamlit run app.py.
```

The dependencies are listed in `../Streamlit/requirements.txt`. The data inputs used by the app are included in the same folder, including `../Streamlit/Data_CMH.csv` and the geographic files used for regional calibration.

S3. ECONOMIC STATE, ACTIONS, AND TYPES

At date t , the public state is summarized by

$$X_t = (t, \mu_t, z_t, \theta_V, \alpha_t, \gamma_t, \mathcal{H}_t),$$

where μ_t is the public posterior over criminal types, z_t are regional and case characteristics, θ_V denotes victim characteristics, α_t is the financial blocking instrument, γ_t is operational pressure, and $\mathcal{H}_t$ is the public history. The latent type is $\theta_K \in \Theta_K$.

For each agent $i \in \{K, S, F\}$, the intended action is $a_t^{i*} \in \mathcal{A}^i$. The application distinguishes intended actions from executed actions:

$$\tilde{a}_t^i \sim \Pi^i(\cdot \mid a_t^{i*}, X_t).$$

The executed profile

$$\tilde{A}_t = (\tilde{a}_t^K, \tilde{a}_t^S, \tilde{a}_t^F)$$

TABLE I
MAIN INPUTS IN THE STREAMLIT CALIBRATION MODEL

Input group	Economic role
Criminal type prior	Initial belief $\mu_0(\theta_K)$, constructed from type baselines, regional information, and victim characteristics.
Victim and region covariates	Observable case characteristics entering the prior, calibration filters, and descriptive diagnostics.
State instruments	Financial blocking α_t and operational pressure γ_t , both chosen inside the policy and mechanism blocks.
Maturity parameter	T_{mad} , which governs the time profile of institutional maturity through $M(t) = \min\{1, (t/T_{\text{mad}})^2\}$.
Residual closure	λ_4 , an exogenous residual closing intensity used in the empirical calibration but separated from the strategic identification block.
Detection parameters	Type-specific intercepts $\eta_0(\theta_K)$ and slopes (η_1, η_2) determining the probability of a detection signal.
Implementation parameters	Temperature and floor parameters governing the MDG implementation kernel from intended to executed actions.
Random seed and horizon	Simulation controls that make the dynamic cycles reproducible for a fixed calibration.

is the profile that enters the public record, the physical-risk block, and the Bayesian updating step.

S4. INPUT LAYER

The calibrated model uses three classes of inputs: case inputs, structural parameters, and simulation controls.

S5. CORE FUNCTIONAL BLOCKS

S5.1. *Competing Risks*

For each criminal type θ_K and executed action profile $\tilde{A}_t$, the app computes four physical intensities:

$$\left(\tilde{\lambda}_1(t | \theta_K), \tilde{\lambda}_2(t | \theta_K), \tilde{\lambda}_3(t | \theta_K), \lambda_4 \right).$$

The first three risks are strategic: payment, kill, and rescue. The fourth risk is a residual exogenous closing channel. Conditional on continuation until date t , the probability of no terminal event during the period is

$$p_{\text{Cont}, t | \theta_K} = \exp \left\{ -\Delta t \sum_{j=1}^4 \tilde{\lambda}_j(t | \theta_K) \right\},$$

with $\tilde{\lambda}_4(t | \theta_K) = \lambda_4$. If a terminal event occurs, the cause probability is proportional to its intensity:

$$\bar{h}_j(t | \theta_K) = (1 - p_{\text{Cont}, t | \theta_K}) \frac{\tilde{\lambda}_j(t | \theta_K)}{\sum_{\ell=1}^4 \tilde{\lambda}_\ell(t | \theta_K)}.$$

Thus the physical outcome is a five-cell categorical variable:

$$m_t \in \{\text{cont}, \text{pay}, \text{kill}, \text{res}, \text{rel}\}.$$

S5.2. Detection Signal

The public detection signal is binary, $d_t \in \{0, 1\}$. Conditional on type and state instruments, its probability is

$$p_{\text{det},t}(\theta_K) = \Lambda(\eta_0(\theta_K) + \eta_1\alpha_t^* + \eta_2\gamma_t^*),$$

where $\Lambda(x) = (1 + \exp\{-x\})^{-1}$. The type-specific intercept captures baseline detectability. The slopes capture the marginal effect of financial blocking and operational pressure.

S5.3. Implementation Kernel

The MDG process maps intended actions into executed actions. For each $i \in \{K, S, F\}$ and $a \in \mathcal{A}^i$,

$$\Pi^i(\tilde{a}_t^i = a \mid a_t^{i*}, X_t) = \frac{\exp(\mathbf{1}\{a = a_t^{i*}\}/T_t)}{\sum_{a' \in \mathcal{A}^i} \exp(\mathbf{1}\{a' = a_t^{i*}\}/T_t)}.$$

The temperature is

$$T_t = T_0 \max \left\{ \frac{H(\mu_t)}{H(\mu_0)} \exp(-\eta_{\text{cal}} t), \underline{c} \right\}.$$

When $\underline{c} > 0$, the probability of executing the intended action remains bounded away from one in the limit:

$$\lim_{t \rightarrow \infty} \Pi^i(\tilde{a}_t^i = a_t^{i*} \mid a_t^{i*}, X_t) = \frac{\exp\{1/(T_0 \underline{c})\}}{\exp\{1/(T_0 \underline{c})\} + |\mathcal{A}^i| - 1} < 1.$$

The special case $\underline{c} = 0$ removes residual implementation noise in the limit and permits full convergence to the intended action.

S5.4. Bayesian Updating

After observing the physical outcome m_t , detection signal d_t , and any additional public signal used by the calibration, the posterior is updated by

$$\mu_{t+1}(\theta_K) = \frac{\mu_t(\theta_K) L_H(m_t \mid \theta_K, X_t) L_D(d_t \mid \theta_K, X_t) L_V(v_t \mid \theta_K, X_t)}{\sum_{\theta' \in \Theta_K} \mu_t(\theta') L_H(m_t \mid \theta', X_t) L_D(d_t \mid \theta', X_t) L_V(v_t \mid \theta', X_t)}.$$

The physical likelihood L_H is generated by the competing-risk block. The detection likelihood L_D is generated by the logistic signal. The optional voice or public-signal likelihood L_V is generated by the calibrated signal block in the application.

TABLE II
MAIN MODEL OUTPUTS

Output	Economic interpretation
Posterior path $\{\mu_t\}$	Public learning over criminal types.
Entropy path $H(\mu_t)$	Speed and concentration of informational learning.
Policy path (α_t, γ_t)	State financial blocking and operational pressure chosen by the mechanism.
Action path	Intended and executed actions for the criminal organization, state, and family.
Physical probabilities	Probabilities of continuation, payment, death, rescue, and residual release.
Detection probabilities	Type-specific and posterior-averaged probability of observing the detection signal.
Incentive diagnostics	Reported IR, IC, and enforcement margins used to evaluate feasibility of the calibrated mechanism.
Exported report	A reproducible summary of the calibration state and dynamic cycle outcomes.

S6. CHOICE AND MECHANISM CALCULATIONS

The app computes the state and family choices from the calibrated structural environment. The state problem evaluates feasible combinations of (α_t, γ_t) and selects the policy profile that minimizes the calibrated expected objective while respecting the implemented feasibility and incentive constraints. The family and criminal organization blocks compute their corresponding payoff-relevant actions given the posterior, the state policy, and the expected continuation values.

The dynamic mechanism therefore has the following sequence in each cycle:

$$\mu_t \longrightarrow (a_t^{K*}, a_t^{S*}, a_t^{F*}) \longrightarrow \tilde{A}_t \longrightarrow (m_t, d_t, v_t) \longrightarrow \mu_{t+1}.$$

S7. TABLE 5.2 WORKFLOW

Table 5.2 is generated by the dynamic cycle engine in the app. At the initial cycle the model computes the baseline posterior, the optimal policy instruments, the intended actions, the implementation probabilities, and the physical outcome probabilities. In subsequent cycles the app repeats the same decision and updating steps using the posterior inherited from the previous cycle.

For a fixed seed and fixed calibration, the simulated history is reproducible. If the realized outcome is different from continuation, the stochastic trajectory terminates in the simulation mode that uses realized draws. In deterministic or diagnostic displays, the app can still report the expected probability vector over the five physical outcomes.

S8. TAB 6 DYNAMIC OUTPUTS

Tab 6 reports dynamic objects derived from the same cycle table used by the calibration engine. The main outputs are:

S9. CODE MAP

S10. REPRODUCIBILITY AND INVARIANCE

The app uses deterministic calculations conditional on the selected inputs. Any stochastic cycle is reproducible once the seed, horizon, and calibration parameters are fixed. Performance

TABLE III
COMPUTATIONAL MAP OF THE STREAMLIT IMPLEMENTATION

File or function block	Role in the economic model
<code>app.py</code>	User-facing calibration engine, state construction, dynamic cycle execution, Table 5.2, and Tab 6 diagnostics.
<code>model_logic.py</code>	Structural payoff and hazard calculations used to map actions and type into intensities and expected values.
<code>rational_behavior.py</code>	Behavioral rules, posterior calculations, mechanism restrictions, and rational-action tables.
<code>dynamic_report.py</code>	Construction of the reproducible dynamic report from the current calibration state.
<code>export_calibration_results.py</code>	Export routines for calibrated outputs.
<code>Data_CMH.csv</code>	Empirical case input used by the calibration interface.

features in the application, such as cached data loading and lazy rendering of expensive figures, do not alter the economic calculations. They only change when intermediate objects are recomputed or displayed.

Consequently, the same calibration state generates the same Table 5.2 values and the same Tab 6 dynamic objects whether the app is run interactively or used as a replication package.